

JEE MAIN REVISION MASTER: ROTATIONAL MOTION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 247 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Moment of Inertia & Parallel/Perpendicular Axes Theorems	208 PYQs	Inertia
Very High	Torque, Angular Momentum & Pure Rolling	109 PYQs	Dynamics & Rolling

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Moment of Inertia & Parallel/Perpendicular Axes Theorems	$I = \int r^2 dm$, $I = I_{\text{cm}} + Md^2$, $I_z = I_x + I_y$ ($\text{laminar planar bodies}$)
Torque, Angular Momentum & Pure Rolling	$\vec{\tau} = I\vec{\alpha} = \frac{d\vec{L}}{dt}$, $K_{\text{roll}} = \frac{1}{2}Mv_{\text{cm}}^2 \left(1 + \frac{k^2}{R^2}\right)$, $a = \frac{g \sin \theta}{1 + k^2/R^2}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Moment of Inertia & Parallel/Perpendicular Axes Theorems (Very High Yield — 208 PYQs)

Core Principle: Calculation of rotational inertia, radius of gyration $k = \sqrt{I/M}$, and axes shift theorems.

■ **Caution / JEE Pitfall:** Perpendicular axis theorem $I_z = I_x + I_y$ is strictly valid ONLY for 2D laminar (flat) objects lying in the xy-plane.

■ **Speed Trick:** Solid sphere $I_{\text{cm}} = (2/5)MR^2$; hollow spherical shell $I_{\text{cm}} = (2/3)MR^2$; circular disc $I_{\text{cm}} = (1/2)MR^2$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A solid cylinder having radius R and length L is slipping on a rough horizontal plane. At time $t=0$ the cylinder has a translational ve..." [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "An object of uniform density rolls up the curved path with the initial velocity v_{o} as shown in the figure. If the maximum height attain..." [Ans: D]
- [2026 Morning Shift] Q#1 (MCQ): "Two circular discs of radius each 10 cm are joined at their centres by a rod of length 30 cm and mass 600 gm as shown in figure. If the mass..." [Ans: D]

3.2 Torque, Angular Momentum & Pure Rolling (Very High Yield — 109 PYQs)

Core Principle: Rotational analogue of Newton's second law, conservation of angular momentum, and rolling down an incline.

■ **Caution / JEE Pitfall:** On an incline, the body with the smallest k^2/R^2 reaches the bottom first (Solid Sphere > Solid Cylinder > Hollow Sphere > Ring).

■ **Speed Trick:** In pure rolling on a flat surface, the instantaneous axis of rotation passes through the contact point ($v_{\text{contact}} = 0$).

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4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A solid cylinder having radius R and length L is slipping on a rough horizontal plane. At time $t=0$ the cylinder has a translational velocity $v_o=49\text{m/s}$, perpendicular to its axis and a rotational velocity $\omega_o/4R$ about the centre. The time taken by the cylinder to start rolling is ____ seconds. (coefficient of kinetic friction $\mu_k=0.25$ and $g=9.8\text{m/s}^2$)

- (A) 15
- (B) 5 **[Correct]**
- (C) 10
- (D) 7.5

Correct Answer: B

Solution Analysis: The cylinder is slipping, which means its translational motion and rotational motion are not in order ($v=r\omega$) for pure rolling. Initial translational velocity is $v_o=49\text{m/s}$. Initial angular velocity is $\omega_o=\frac{v_o}{4R}$, where R is the radius of solid cylinder of length L . For pure rolling to start, the condition $v=\omega R$ must be satisfied. Initially, $\omega_o R=\frac{v_o}{4}$, which is less than v_o , so the cylinder is sliding forward. Since the cylinder is sliding forward, kinetic friction (f_k) acts backward at the point of contact. $f_k=\mu_k$

Benchmark Problem 2: JEE Main 2026 (Online) 2nd April Morning Shift (Q#3)

The position of an object having mass 0.1 kg as a function of time t is given as $\vec{r}(t) = (10t^2\hat{i} + 5t^3\hat{j})$ m. At $t = 1$ s, which of the following statements are correct ? A. The linear momentum $\vec{p} = (2\hat{i} + 1.5\hat{j})$ kg·m/s. B. The force acting on the object $\vec{F} = (2\hat{i} + 3\hat{j})$ N. C. The angular momentum of the object about its origin $\vec{L} = 15k$ J·s. D. The torque acting on the object about its origin $\vec{\tau} = 20k$ N·m. Choose the correct answer from the options given below:

- (A) , B and C only
- (B) , C and D only
- (C) , C and D only
- (D) , B and D only **[Correct]**

Correct Answer: D

Solution Analysis: The mass of object is $m = 0.1$ kg. The position of the object is given as $\vec{r}(t) = 10t^2\hat{i} + 5t^3\hat{j}$. We evaluate each statement at $t = 1$ s. Linear momentum is defined as $\vec{p} = m\vec{v}$. The velocity is defined as the rate of change of position, $\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(10t^2\hat{i} + 5t^3\hat{j}) = 20t\hat{i} + 15t^2\hat{j}$. At $t = 1$ s : $\vec{v} = 20(1)\hat{i} + 15(1)^2\hat{j} = (20\hat{i} + 15\hat{j})$ m/s. So, the linear momentum of the object is, $\vec{p} = 0.1 \times (20\hat{i} + 15\hat{j}) = (2\hat{i} + 1.5\hat{j})$ kg · m/s. Thus,